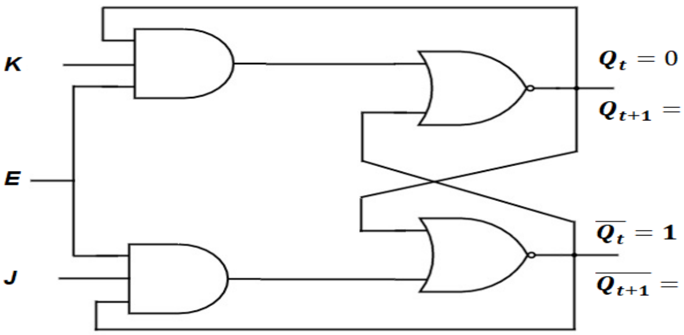
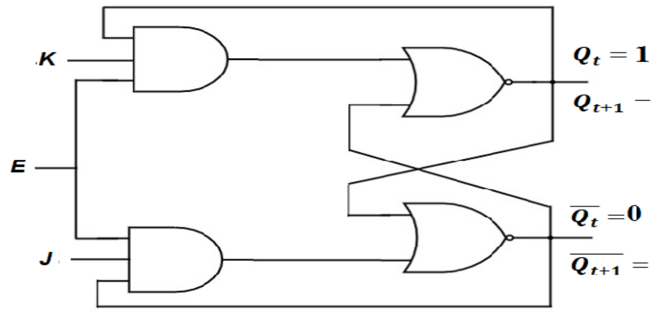
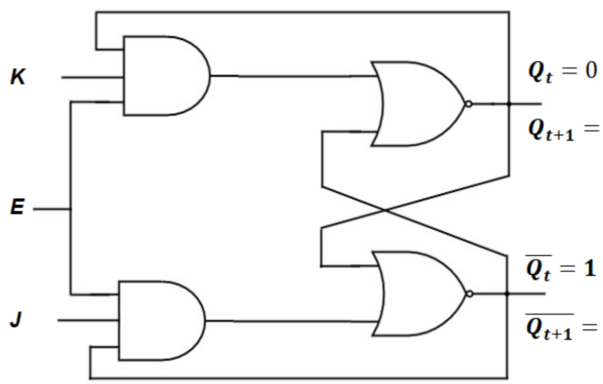
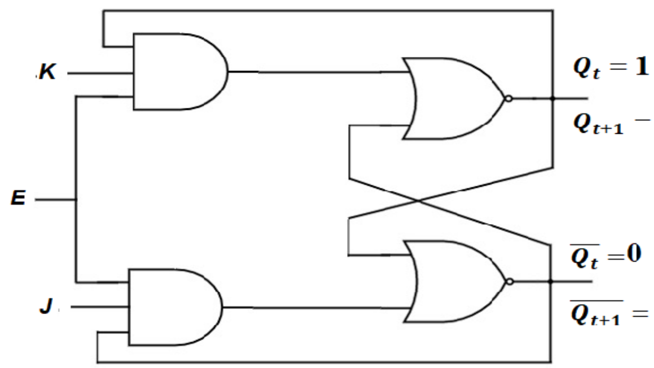
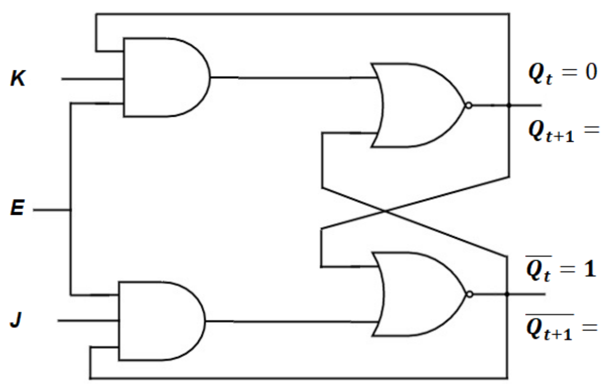
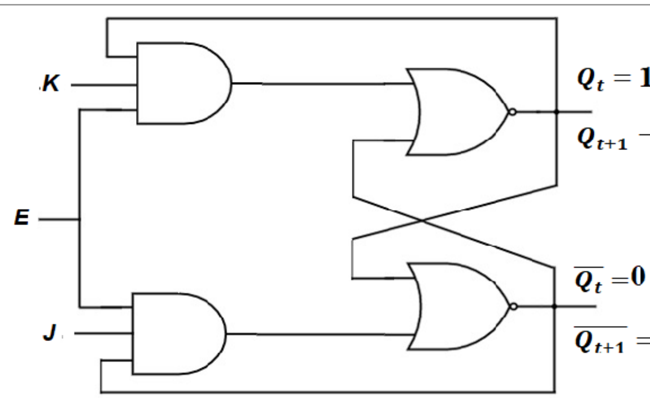
QUIZ 7-A

ID:________ NAME:________________________________ SECTION: ____

Instruction: **No marks will be awarded for Direct Answers**

1. *Process the **JK Latch** given below for all possible cases and fill the table accordingly?*

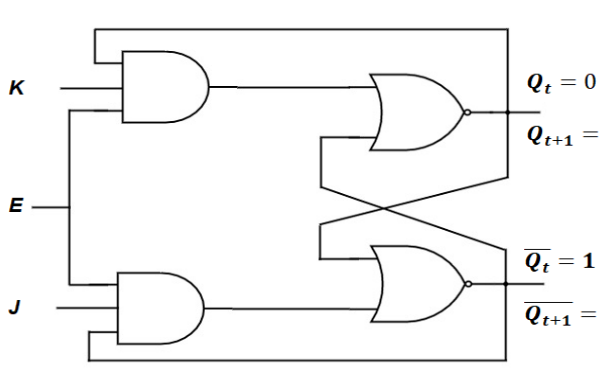
CIRCUIT:

Case-I ($Q_t = 0$)	**Case-II ($Q_t = 1$)**
E=0, J= , K= , $Q_t = 0$, Q_{t+1} = ___	**E=0, J= , K= , $Q_t = 1$, Q_{t+1} = ___**
	
E=1, J=0 , K=0 , $Q_t = 0$, Q_{t+1} = ____	**E=1, J=0 , K=0 , $Q_t = 1$, Q_{t+1} =____**
	
E=1, J=0 , K=1 , $Q_t = 0$, Q_{t+1} =_____	**E=1, J=0 , K=1 , $Q_t = 1$, Q_{t+1} =____**
	

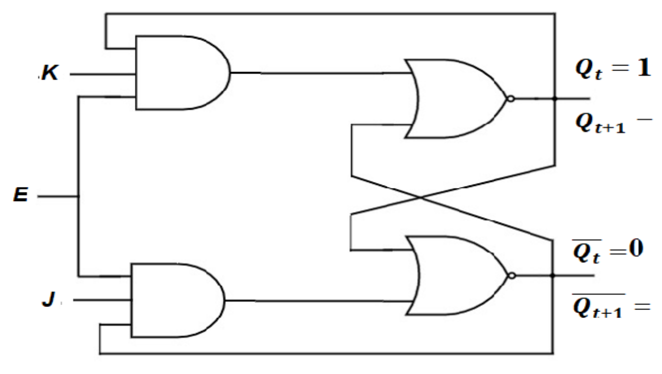
QUIZ 7-A

ID:________ NAME:___________________________________ SECTION: ____

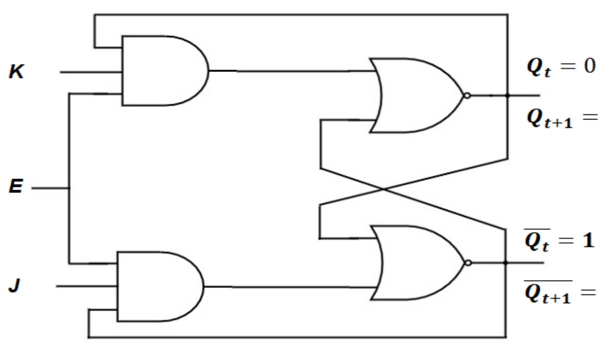
E=1, J=1, K=0 , $Q_t = 0$, Q_{t+1} =_____



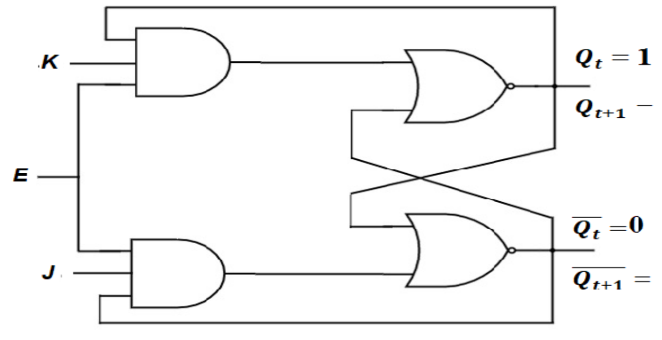
E=1, J=1, K=0 , $Q_t = 1$, Q_{t+1} =____



E=1, J=1 , K=1 , $Q_t = 0$, Q_{t+1} = ____



E=1, J=1 , K=1 , $Q_t = 1$, Q_{t+1} = ___



JK Latch

E	J	K	Q_{t+1}	$\overline{Q_{t+1}}$
0	X	X		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

JK LATCH with SET/RESET

SET	CLEAR	Q_{t+1}	$\overline{Q}_{t+1}$
0	0		
0	1		
1	0		
1	1		

2. *Modify given NOR Based JK Flip Flop circuit to introduce Asynchronous **CLEAR** and **SET** inputs fill table above?*